1 Blood pressure in the arteries changes during the cardiac cycle. The highest blood pressure occurs during ventricular systole when the ventricles contract and push blood out of the heart. The lowest blood pressure occurs during diastole when the atria and the ventricles are relaxed.

Blood pressure is measured as two values:

- systolic blood pressure
- diastolic blood pressure.

A blood pressure monitor, as shown in Figure 1.1, is used to measure the blood pressure in the brachial artery in the arm.

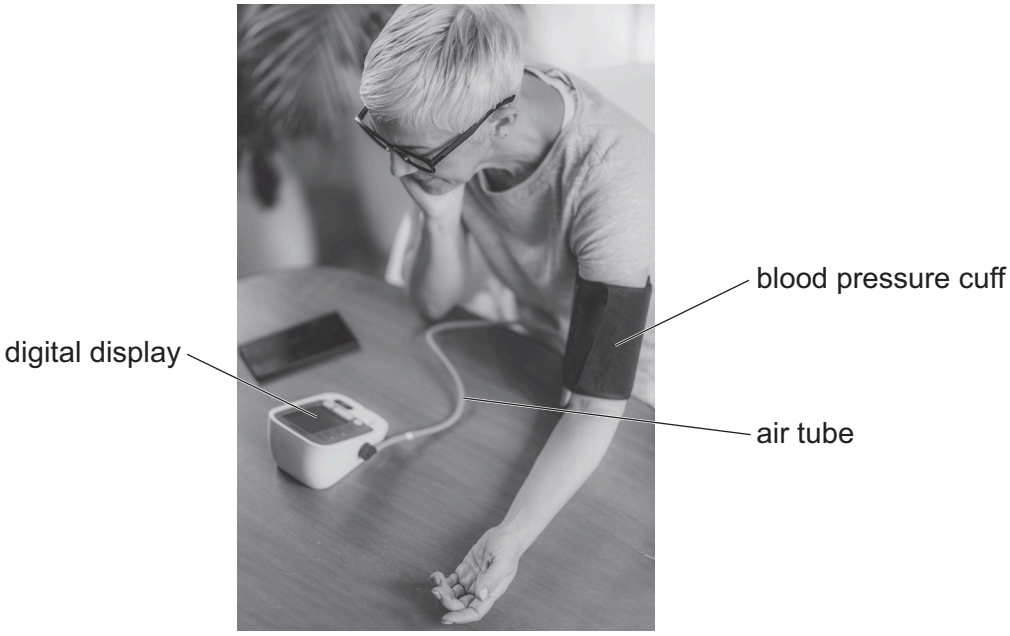

Figure 1.1

Some scientists investigated the effect of a 12-week exercise programme on the blood pressure of a group of 18 people. The people were aged from 40 years to 64 years, did **not** exercise regularly and had jobs that mainly involved sitting.

The exercise programme consisted of a sequence of three activities, repeated three times a week for 12 weeks.

The three activities in each sequence were:

- walking for 5 minutes
- running slowly for 30 minutes
- walking for 5 minutes.

(a) The time spent walking and running, the number of repeats and the 12-week period were all standardised.

Identify **two other** variables in this exercise programme that the scientists should standardise.

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..... [2]

(b) On the morning **before** the start of the 12-week exercise programme, blood pressure measurements were taken for the group of 18 people, as shown in Figure 1.2. Every blood pressure measurement included a systolic and a diastolic blood pressure reading.

1 Resting blood pressure was measured while each person was sitting.

2 Each person then carried out a walking test on a treadmill for 15 minutes.The gradient of the treadmill was increased every minute.

3 After the walking test, the mean blood pressure of each person was determined from blood pressure measurements taken every 20 minutes over a period of 10 hours.

4 For each person, the scientists calculated the **difference** between the mean blood pressure after the walking test and the resting blood pressure.

$$\begin{pmatrix}\text{difference}\\ \text{in blood}\\ \text{pressure}\end{pmatrix} = \begin{pmatrix}\text{mean blood}\\ \text{pressure after}\\ \text{the walking test}\end{pmatrix} - \begin{pmatrix}\text{resting blood}\\ \text{pressure before}\\ \text{the walking test}\end{pmatrix}$$

5 The **mean** difference in blood pressure was then calculated for the 18 people in the study.

Figure 1.2

The same blood pressure measurements were taken on the morning **after** the end of the 12-week exercise programme and used to recalculate the mean difference in blood pressure for the 18 people in the study.

Figure 1.3 shows the results of this investigation.

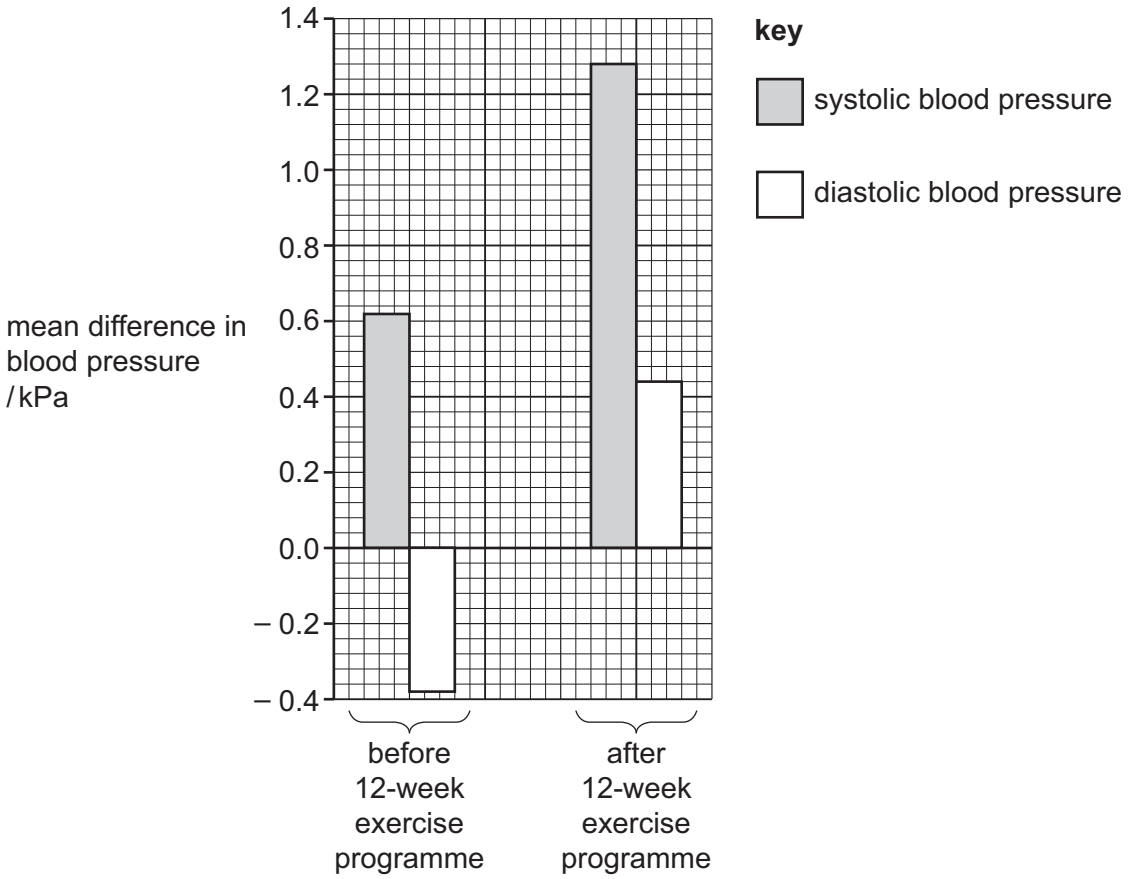

Figure 1.3

(i) Identify the **independent** variable in this investigation.

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..... [1]

(ii) Describe the results of the investigation shown in Figure 1.3.

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..... [3]

(c) Table 1.1 shows the mean resting systolic blood pressure data for the 18 people included in the investigation.

Table 1.1

before or after 12-week exercise programme	mean resting systolic blood pressure /kPa	sample standard deviation /kPa	standard error (SE) /kPa
before	17.71	1.33	.....
after	17.08	1.35	0.32

(i) The formula for the calculation of standard error (SE) is:

$$SE = \frac{s}{\sqrt{n}}$$

key to symbols

s = sample standard deviation

n = sample size (number of observations)

Complete Table 1.1 by calculating the SE for the mean resting systolic blood pressure before the 12-week exercise programme. [1]

(ii) Suggest why it is helpful to calculate the standard error for the mean resting systolic blood pressure before and after the 12-week exercise programme.

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.....

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..... [2]

[Total: 9]